**Inorganic Cumulative Exam**

**September 11, 2015**

Visible absorption spectroscopy is a fundamental tool in traditional coordination chemistry and solid state chemistry. The following questions are typical examples of applications. 20 pts each.

1. $[Co(NH_3)_4Cl_2]Cl$ can be made in two forms one is purple the other yellow. Explain the source of this difference in color.
2. Emeralds are green yet the chemical formula is often written as $Be_3Al_2(SiO_3)_6$ which is the same as the mineral beryl which is colorless. Rubies which are red and can likewise be formulated as colorless aluminum oxide. Both rubies and emeralds exhibit absorption spectra very similar to transition metal complexes dissolved in a solvent. Rationalized the source of the color.
3. Unlike classic coordination compounds the absorption spectra of quantum dots are often discussed but rarely in terms of the chemical composition. Why? CdS is a common example and it is a $d^{10}$ system.
4. Some transition metal complexes show intense charge transfer absorption bands that shift significantly (red to green) with the choice of solvent in which the spectra is obtained. Provide a reasonable hypothesis for this observation. The answer you provide will be judged on the scientific soundness of your hypothesis and not how well it fits to current models. Hint: The fact that the band is due to charge transfer is very important.
5. Many colored glasses used by glass blowers changes color dramatically when heated but return to the original color on cooling. For example yellow glass turns deep red on heating. Provide a reasonable hypothesis for this observation. The answer you provide will be judged on the scientific soundness of your hypothesis and not how well it fits to current models.